

Customer Order And Review Analysis Using Python

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INTRODUCTION:

The project uses two datasets—Customer data and Order data—which are merged using a common key “CustomerID” to create a new dataset for analysis. After merging a new column called Revenue is calculated by multiplying product price and quantity.

Various analytical operations are performed, such as:

- 1) Revenue analysis by category, city, and customer.
- 2) Identification of monthly order trends.
- 3) Exploration of data distribution and patterns.

To understanding, the project includes data visualizations using Matplotlib and Seaborn:

- 1) Bar charts for revenue comparison
- 2) Count plots for order distribution
- 3) Line graphs for time-based trends
- 4) Histograms for quantity distribution

INSTALL PACKAGE:

- 1) pip install numpy
- 2) pip install pandas
- 3) pip install matplotlib
- 4) pip install seaborn

CODE IMPLEMENTATION:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Dataset
```

```
df_Customer = pd.read_csv(r"C:\Users\DHRUV  
SHAH\Customer Order & Review\Customer.csv")  
df_Order = pd.read_csv(r"C:\Users\DHRUV SHAH\Customer  
Order & Review\Order.csv")  
print(df_Customer.head())  
print()  
print(df_Order.head())  
print()  
  
# Merge two dataset  
merged_df = pd.merge(df_Customer, df_Order,  
on="CustomerID", how="outer")  
print(merged_df)  
print()  
  
# Add Revenue column  
merged_df['Revenue'] = merged_df['Price'] *  
merged_df['Quantity']  
print(merged_df)  
print()  
  
# Revenue by Category  
Revenue_Category =  
merged_df.groupby('Category')['Revenue'].sum()  
print(Revenue_Category)  
print()  
  
# Revenue by City  
Revenue_City =  
merged_df.groupby('City')['Revenue'].sum()  
print(Revenue_City)  
print()  
  
# Revenue by Customer
```

```
Revenue_CustomerName =
merged_df.groupby('CustomerName')['Revenue'].sum()
print(Revenue_CustomerName)
print()

# Monthly Trend
merged_df["OrderDate"] =
pd.to_datetime(merged_df["OrderDate"])
merged_df["Month"] =
merged_df["OrderDate"].dt.to_period("M")
monthly_orders =
merged_df.groupby("Month")["OrderID"].count().reset_in
dex()
print(monthly_orders)
print()

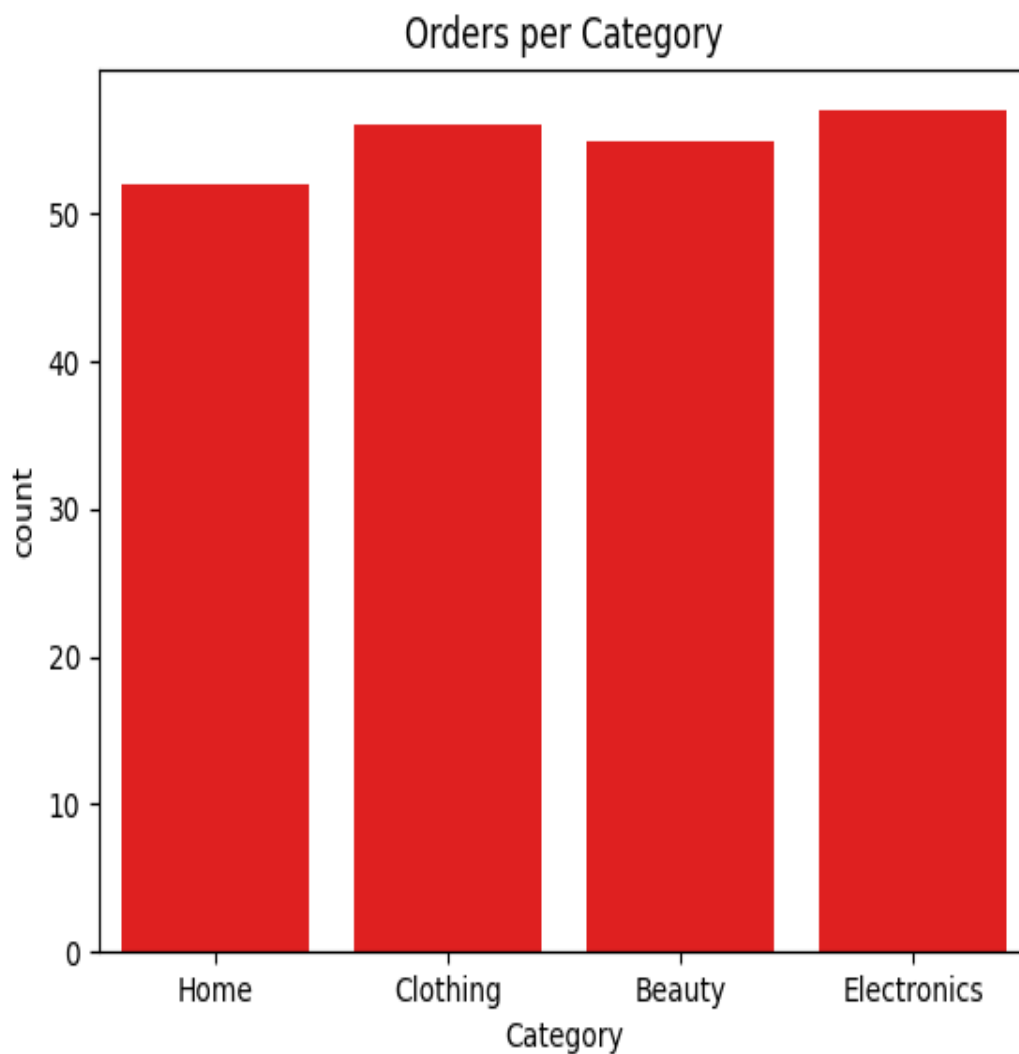
# Visualization
sns.countplot(x="Category", data=merged_df, color="r")
plt.title("Orders per Category")
plt.show()
print()

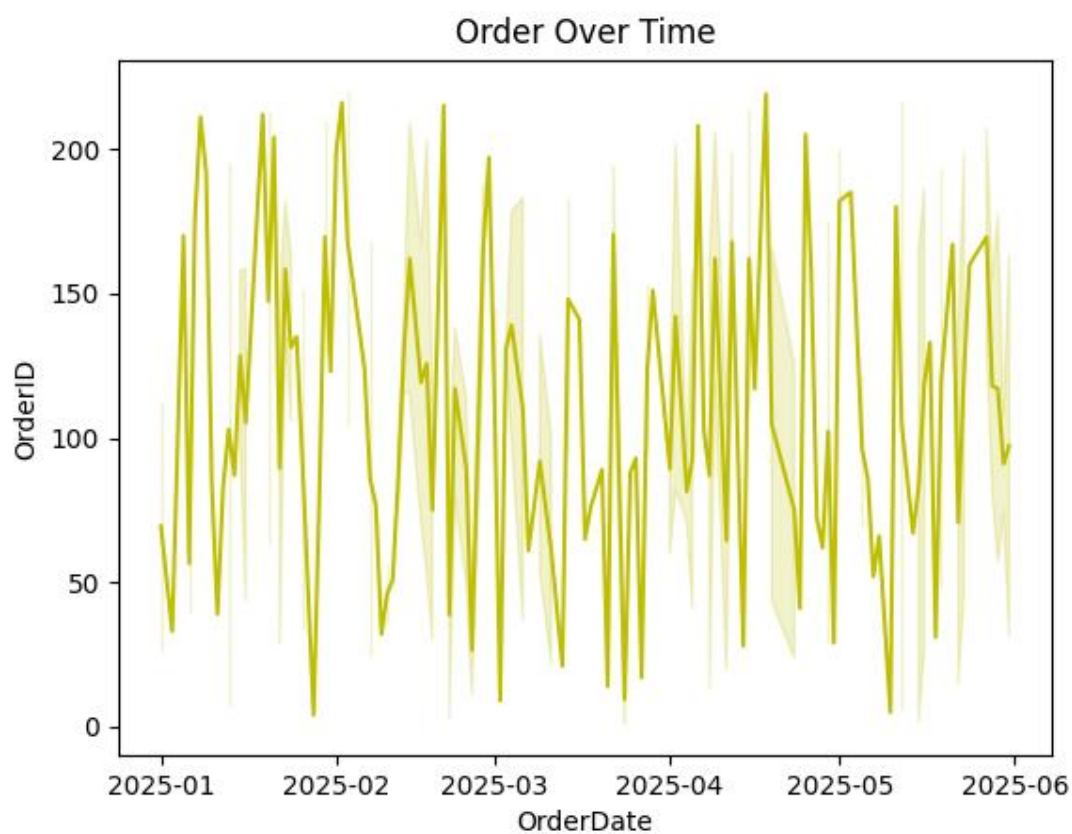
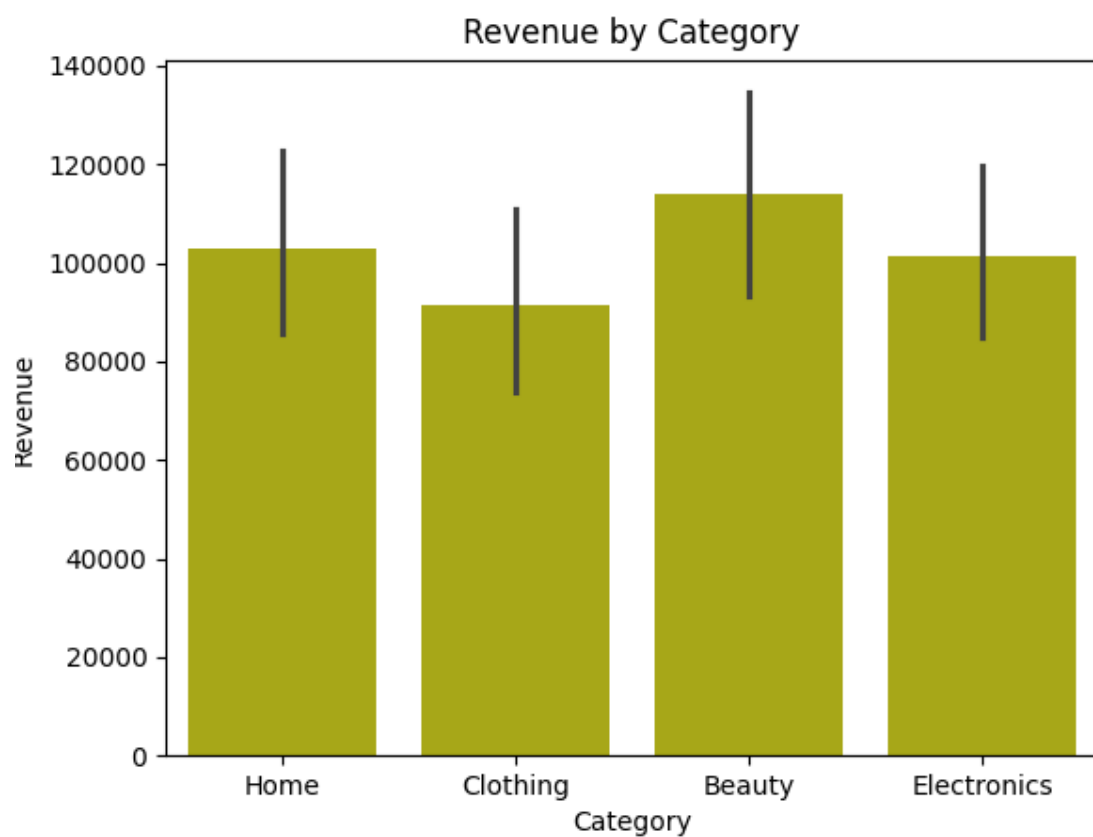
sns.barplot(x="Category", y="Revenue", data=merged_df,
color="y")
plt.title("Revenue by Category")
plt.show()
print()

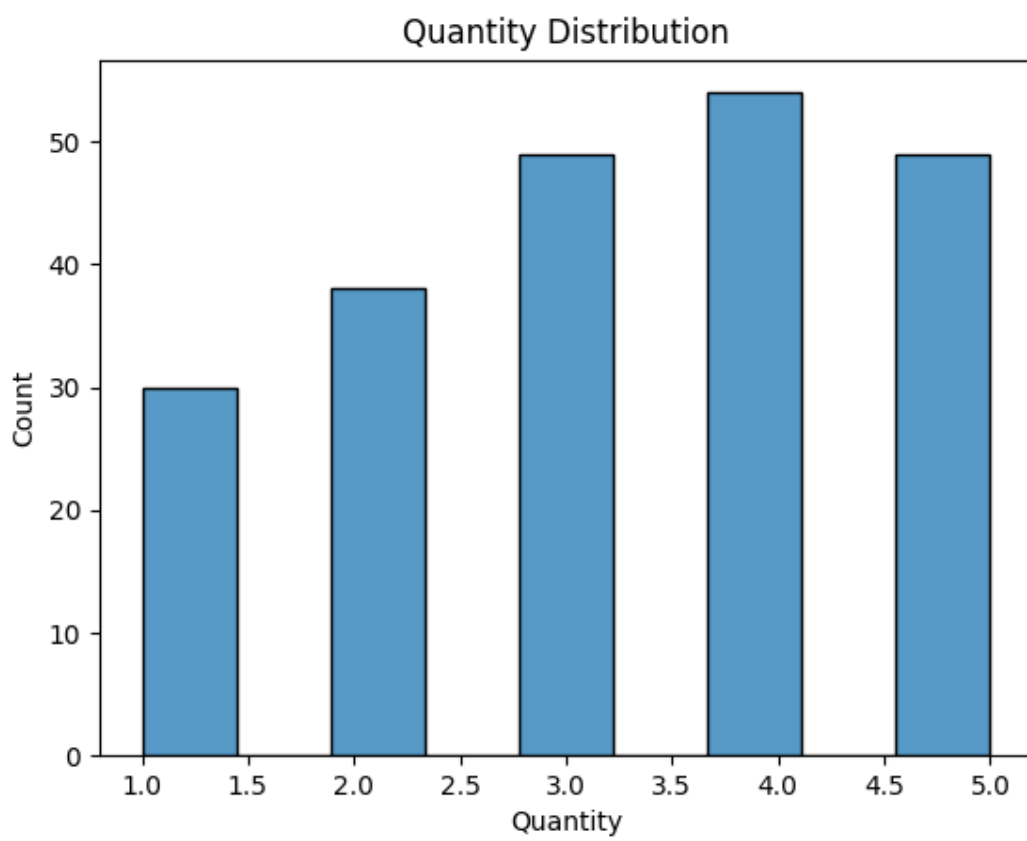
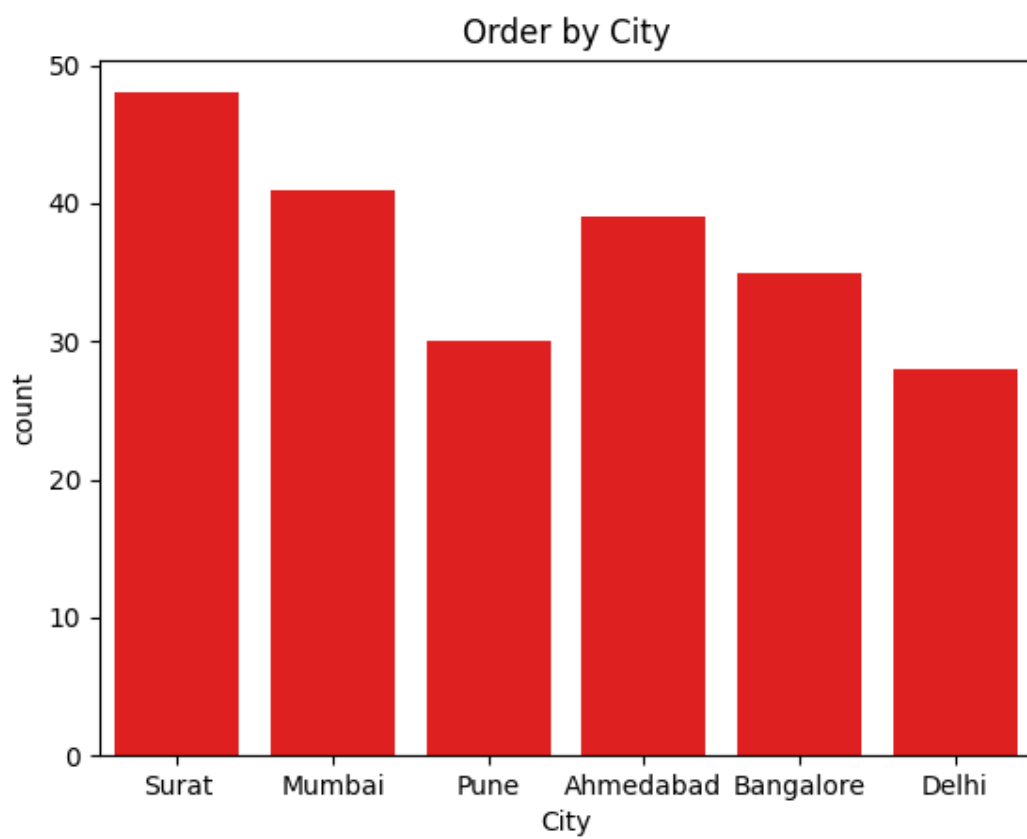
sns.lineplot(x="OrderDate", y="OrderID",
data=merged_df, color="y")
plt.title("Order Over Time")
plt.show()
print()
```

```
sns.countplot(x="City", data=merged_df, color="r")
plt.title("Order by City")
plt.show()
print()

sns.histplot(merged_df["Quantity"])
plt.title("Quantity Distribution")
plt.show()
```







OBSERVATIONS:

- 1) Merging datasets helps in analyzing customer and order details together.
- 2) Revenue is calculated using $\text{Price} \times \text{Quantity}$ to measure total sales.
- 3) Some product categories generate higher revenue than others.
- 4) Most orders have low quantity, while high-quantity orders are rare.